

Functional & Performance Testing Template

Model Performance Test

Date	10 July 2026
Team ID	
Project Name	OptiCrop : Smart Agricultural Production Optimization Engine
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Home Page Landing	Launch the Flask application and open the home page in the browser.	Home page loads successfully with all input fields displayed.	Home page loaded successfully with prediction form.	PASS
FT-02	Input Validation	Enter valid and invalid values for Nitrogen, Phosphorous, Potassium, Temperature, Humidity, pH, and Rainfall.	Accepts valid inputs and displays appropriate error messages for invalid values.	Validation worked correctly for all tested inputs.	PASS
FT-03	Crop Recommendation Prediction	Enter valid agricultural parameters and click Predict.	Recommended crop is displayed along with confidence score and crop information.	Crop recommendation displayed correctly with confidence and crop details.	PASS
FT-04	Backend Model Integration	Submit prediction request and verify communication between Flask backend and trained model.	Flask successfully loads the Pickle model and returns predictions.	Backend successfully loaded the model and generated predictions.	PASS
PT-01	Response Time Test	Measure the time taken to generate a prediction.	Prediction should be generated within 2 seconds on the local machine.	Average response time observed: 0.5–1 second .	PASS

PT-02	Multiple Prediction Requests	Perform several predictions consecutively using different input values.	Application should respond consistently without crashes or errors	Successfully handled multiple prediction requests.	PASS
PT-03	Model Performance Test	Test the application using different samples from the dataset.	Model should provide accurate crop recommendations with high accuracy.	Random Forest achieved 99.32% accuracy with correct recommendations.	PASS